

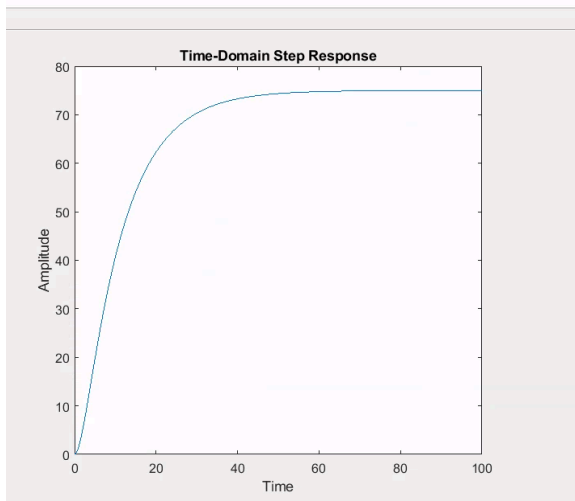
ELEG418 – Course project PID controller design for traffic lights Spring 2024

Gerald Stanford

PART 1: Inspecting The Manipulator

Determining the Step Response of the numerator:

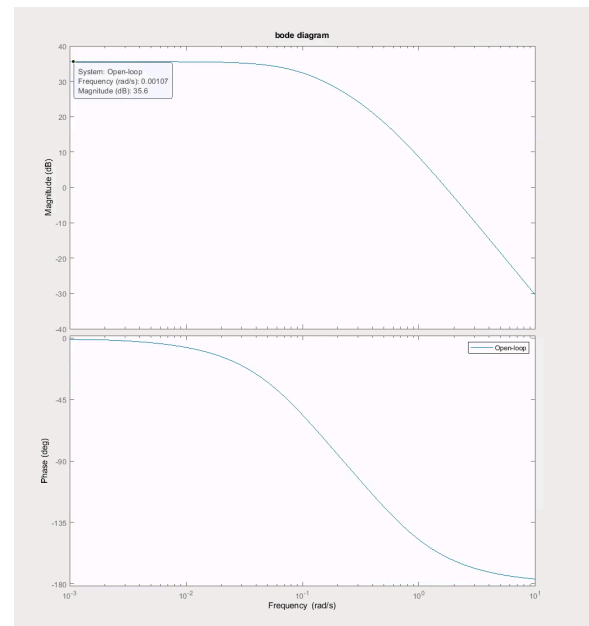
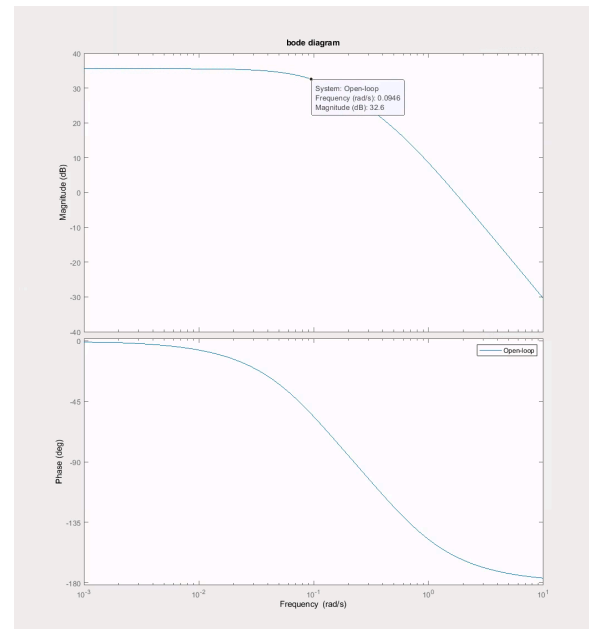
Code Segment used:
Get_transfer_display_poles.m



Conclusions From Step Response

The graph of the step response shows that the gain reaches roughly 75. It is a second order system and the 2 poles must be on the real axis both on the left hand plane of the imaginary axis, but not identical poles. The behavior of the curve is an Overdamped curve. The settling time of the system is about 41 seconds and the system is stable.

Code Segment Used: Plot_bode_transfer.m

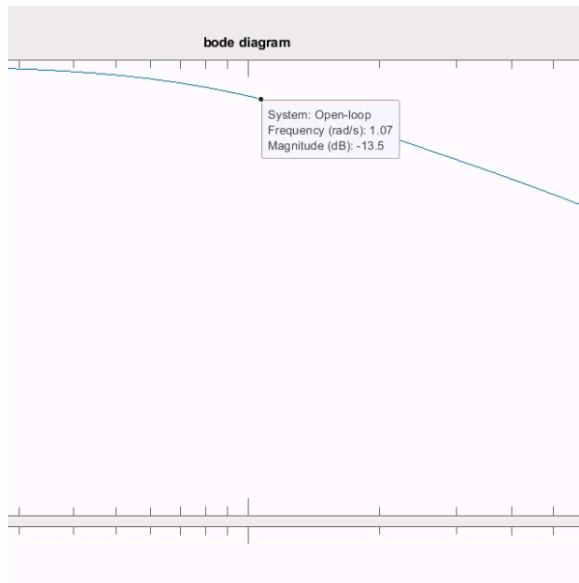


Shows the initial frequency and the corner freq

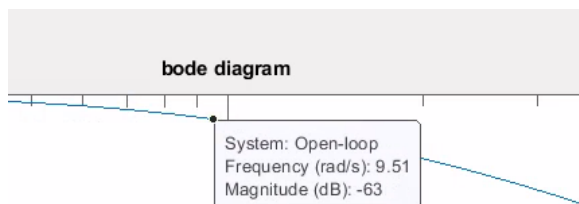
Increasing the Corner Frequency

Code Segment Used: Plot_bode_transfer.m
(sys altered)

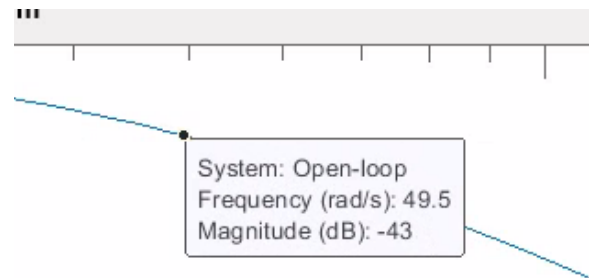
Result of reducing the gain in the numerator



Result of increasing the pole values, up to 10 and 50



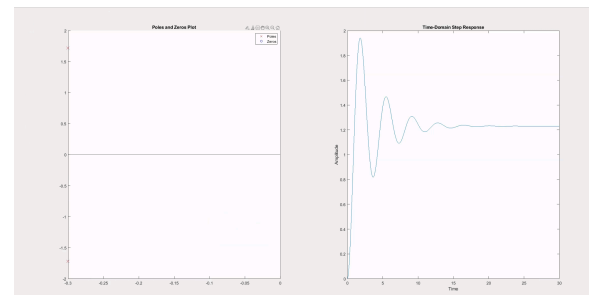
Result of introducing a left hand plane zero in the transfer function



By adding a zero, decreasing gain and moving poles farther left, the I was able to push the corner frequency to a much later frequency than initially.



Closed Loop Response:
Code Segment Used:
Get_feedback_display.m



Closed loop/ Open loop comparison:

- The poles changed positions and no longer on the real axis
- More oscillations in signal
- Was overdamped now underdamped with overshoot
- Gain is significantly lower
- Settling time is shorter

PART 2: Designing PID Controller

Code Segment Used: pid_controller.m ;
pid_controller_ranges.m; pid_plug_in.m

Reasoning Behind Controller choice

Initially I attempted to find a coefficient K_p that would be able to stabilize the system as well as meet the desired component requirements. But in my attempt to do so I realized trying to find a value of K_p that could = K_u the maximum gain was difficult. The poles of the system never reached the imaginary axis and instead moved toward the breakpoint and headed towards infinity upwards and downwards at point -0.3 on the real axis. I also attempted to use the Ziegler-Nichols but the results were undesirable and the system output became unbounded.

I then proceeded to reduce K_p to achieve quarter amplitude decay. Through trial and error of doing the root locus, I found ranges of about 1.2 to 1.8 somewhat suitable. Eventually I settled on values of about 1.75.

After finding a suitable K_p value I tried to find a coefficient K_d that would result in the desired performance overshoot percentage and settling time.

Initially I tried to perform the root locus method again with an updated transfer function, but found the process difficult, perhaps due to lack of understanding. So I instead created a loop to spot the desired results.

In the end having a K_d value of 2 with a k_p value of 1.75, was able to Bring the percentage overshoot to below 3% and keep the settling time under the 3s. So I decided that a PD controller would be sufficient

```
%find kd
for kd = 0:10
    controller = pid(kp,ki,kd)
    sys = 3 / ((s+0.1) * (s + 0.5))
    feedback_sys = feedback(controller * sys, 1);
    %% step(feedback_sys)

    % step data
    s_info = stepinfo(feedback_sys);

    overshoot = s_info.Overshoot;
    settling_time = s_info.SettlingTime;

    fprintf('Proportional gain (kp): %2.3f\n', kp)
    fprintf('Derivative gain (kd): %2.3f\n', kd)
    fprintf('Percentage overshoot: %2.2f%%\n', overshoot)
    fprintf('Settling time: %2.2f s\n\n', settling_time)

    if settling_time <= 3 & overshoot <= 5
        break
    end
end
```

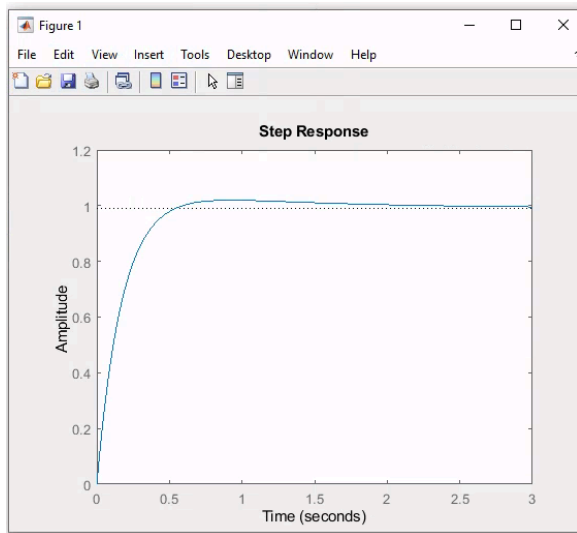
Continuous-time transfer function.

Model Properties

Proportional gain (kp): 1.750
Derivative gain (kd): 2.000
Percentage overshoot: 2.92%
Settling time: 1.51 s

Proportional gain (kp): 1.750
Percentage overshoot: 2.92%
Settling time: 1.51 s

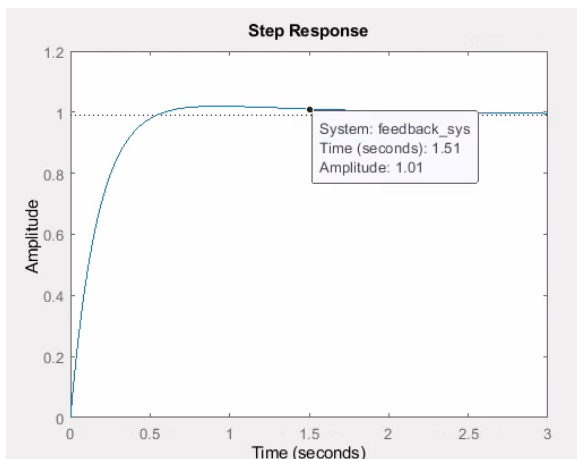




Percent Criterion

Using the current PD controller, assuming the 2 percent is referring to steady state error only $k = 2$: kept the settling time low while also keeping the the steady state error within 2%

$K_d = 2$ with settling time 1.51s and amplitude of 1.01



Conclusions

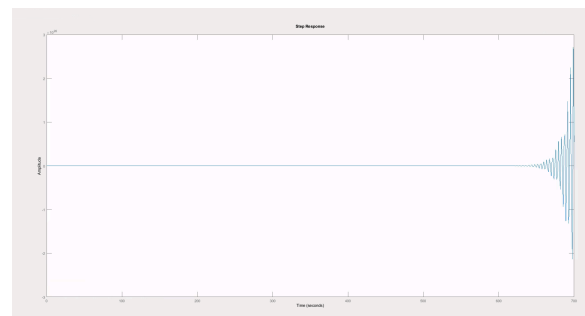
Finding a suitable K_p value is important in determining the K_d and K_i values if they also need to be implemented. However implementing a full PID controller may not also be necessary if the K_p and K_d can be configured to meet the criteria, but using all three seems to provide the system with more flexibility in the range of gains that can be used and could be used if additional criteria are required.

Part 3: Find Values of K

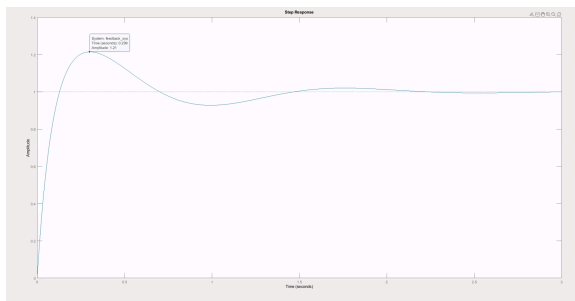
Code Segment Used: Find_K.m

Values of K That Yield Less than 5% overshoot:

The value of 2 results in the system blowing up without bounds.

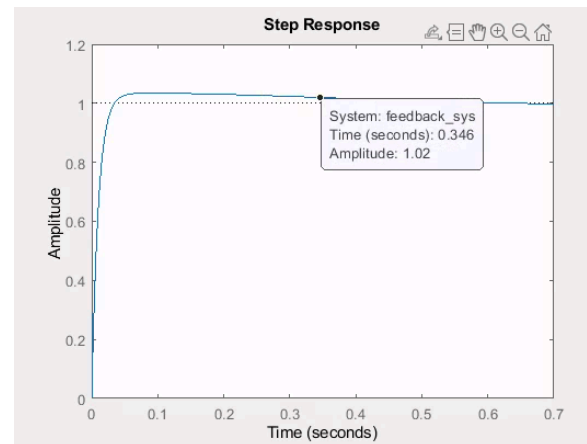


The value of 10 Dampens the system, but yields an overshoot of 21.48.



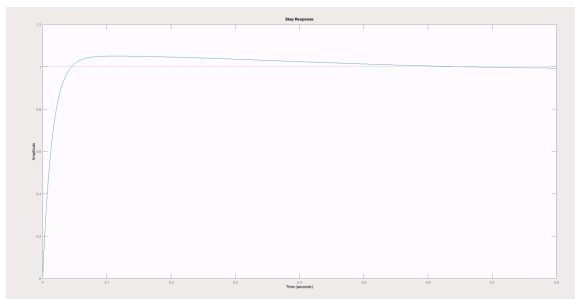
Proportional gain (kp): 2.000
 Derivative gain (kd): 0.500
 Integral gain (ki): 8.000
 Value of K: 68.000
 Percentage overshoot: 3.45%
 Settling time: 0.34 s

Step Response

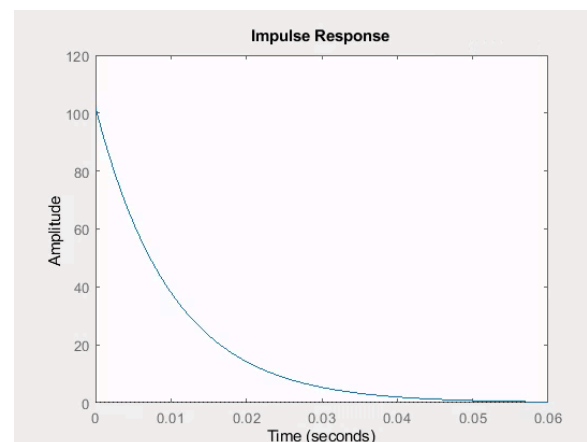


After increasing the value of K and testing the results. K = 47 is the first value to provide an overshoot less than 5.

Proportional gain (kp): 2.000
 Derivative gain (kd): 0.500
 Integral gain (ki): 8.000
 Value of K: 47.000
 Percentage overshoot: 4.99%
 Settling time: 0.43 s



Impulse Response



Minimizing Settling Time

In order to minimize the Settling time and stay within the 2 percent criterion of steady state error, the value of K chosen was 68. It maintained a PO less than 5% and reduced the settling time to 0.34 seconds.

Conclusions

Utilizing a full PID means the controller can be more finely tuned to stabilize the system and providing a scaling factor also increases the range of values suitable to stabilize. Between the two controllers the PD controller was easier to develop but the Second design seems like it could handle a variety of conditions.

That being said I was able to reduce overshoot and decrease the settling time more with my design so based on the traffic light criteria, I would use either in this case.